

Dynamics of Rigid Bodies

Rigid body: collection of particles whose relative distances are fixed.

But! Keep in mind: such bodies actually do not exist as atoms and molecules are vibrating. → WE WILL NEGLECT THIS!!!



→ We are interested in the motion of this system of particles as a whole.

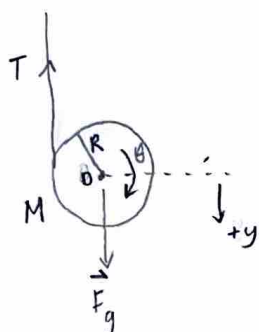
Let us start with an example:

Example:

(Yo-yo problem)

A string attached to the ceiling is wrapped around a homogeneous cylinder.

At $t=0$ the cylinder is dropped from rest and rotates as the string unwinds.



$T = ?$ tension

$\ddot{y} = ?$ linear acceleration (of the center of mass)

$\ddot{\omega} = ?$ angular acceleration

$\omega = ?$ angular velocity (about the cylinder's center)

The center of mass moves $\downarrow$ due to sum of forces:

$$M\ddot{y} = \bar{F}_g - T = Mg - T$$

$$N = RT = -I\ddot{\theta} \rightarrow \text{the torque about O.}$$

↳ rotational inertia of the cylinder ($MR^2/2$)

Initial conditions: $y(0) = \theta(0) = 0$

We know that $y = R\theta$, $\dot{y} = R\dot{\theta}$, $\ddot{y} = R\ddot{\theta}$

So let us combine everything:

$$\ddot{y} = g - \frac{T}{M} = g - \frac{I\ddot{\theta}/R}{M} = g - \frac{MR^2/2 \ddot{\theta}}{MR} = g - \frac{\ddot{y}}{2} \Rightarrow \ddot{y} \left(1 + \frac{1}{2}\right) = g \Rightarrow \ddot{y} = \frac{2}{3}g < g!$$

$$\alpha = \ddot{\theta} = \frac{\ddot{y}}{R} = \frac{2g}{3R} : \text{angular acc.}$$

$$T = \frac{I\ddot{\theta}}{R} = \frac{MR^2}{2} \frac{1}{R} \frac{2g}{3R} = \frac{Mg}{3}$$

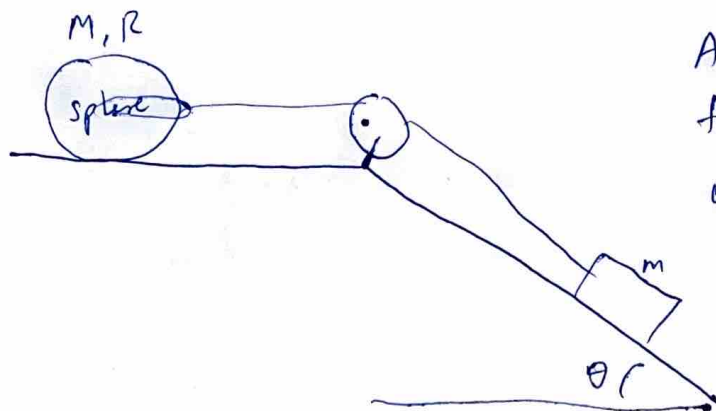
The angular velocity is $\omega = \dot{\theta} = v/R$

$$\int dy = \int \frac{2}{3}g dt \rightarrow \dot{y} = \frac{2}{3}gt = v \rightarrow \omega = \frac{2}{3} \frac{gt}{R}$$

linear acceleration

cm is pulled with an effective force $\frac{2}{3}mg$.

Ex:



A block of mass m is attached to a light string that passes over a light pulley and is connected to a uniform solid sphere of radius R and mass M . Show that the ~~acc.~~ speed of the syst. is

they are the same

$$mgh = \frac{1}{2} M V_{cm}^2 + \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} m v^2$$

$$mgh = \frac{1}{2} M v^2 + \frac{1}{2} \left(\frac{2}{5} M R^2 \right) \left(\frac{v}{R} \right)^2 + \frac{1}{2} m v^2$$

when block is released from rest.

$$a = \frac{g}{1 + \frac{7}{5} \left(\frac{M}{m} \right)}$$

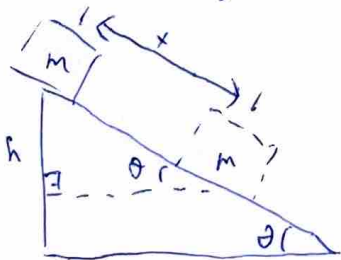
After m goes down a height h

$$2mgh = M v^2 + \frac{2}{5} M \frac{R^2}{R^2} (v^2) + m v^2$$

$$2mgh = v^2 \left(M + \frac{2}{5} M + m \right)$$

$$2mgh = v^2 \left(\frac{7M}{5} + m \right) \rightarrow v = \sqrt{\frac{2mgh}{m + \frac{7M}{5}}} = \sqrt{\frac{2gh}{1 + \frac{7M}{5m}}}$$

If $mgh = \frac{1}{2} m v^2 \rightarrow \sqrt{2gh} = v \Rightarrow \frac{dv}{dt} = \frac{g}{\sqrt{2gh}} \frac{dh}{dt} = \frac{g}{v} \sin \theta = g \sin \theta$



$$\sin \theta = \frac{h}{x}$$

$$h = x \sin \theta$$

$$\frac{dh}{dt} = v \sin \theta$$

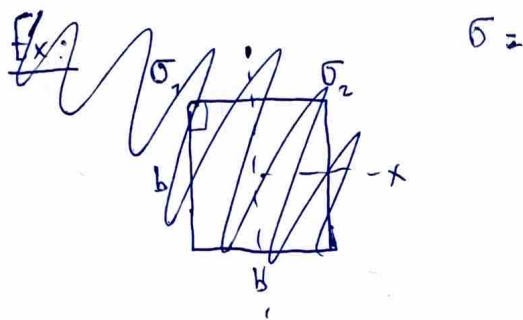
$$\sin \theta = \frac{h}{x} \Rightarrow h = x \sin \theta$$

$$v = \sqrt{2gx \sin \theta}$$

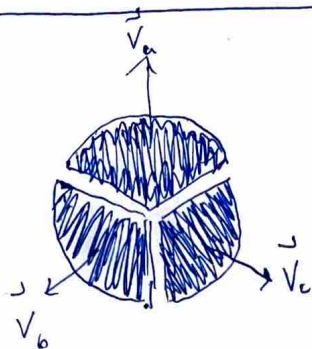
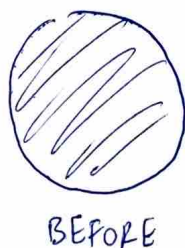
$$\frac{d}{dt} \left(\frac{2mgh}{m + \frac{7M}{5}} \right)^{1/2} = \frac{1}{2} \left(\frac{2mgh}{m + \frac{7M}{5}} \right)^{-1/2} \frac{dh}{dt} \frac{2mg}{2\sqrt{2gh}}$$

$$v = \sqrt{2g \sin \theta x}$$

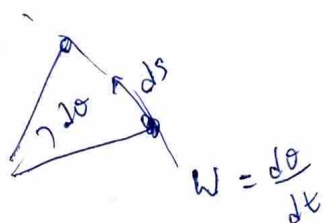
$$\sin \theta = \frac{h}{x} \Rightarrow h = x \sin \theta$$



Ex:



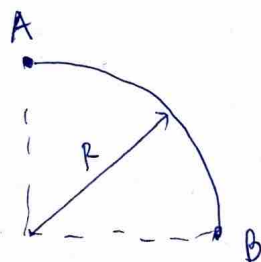
A disc is in rest in the beginning is disintegrated into three completely equal pieces as it is shown in the figure. Each piece have the same mass m .
 $V_a = \frac{2m}{5}$



$$d \frac{ds}{dt} = \frac{d\theta R}{dt} = v$$

$$v = \omega R$$

$$\frac{v}{R} = \omega = \frac{v}{R}$$



Ex: A uniform solid cylinder, sphere and loop roll without slipping from rest at the top of an inclined plane. Find which object would reach the bottom first.

$$I_{cyl} = \frac{1}{2} MR^2 \Rightarrow I/MR^2 = 0.5$$

$$I_{sph} = \frac{2}{5} MR^2 \Rightarrow 0.4$$

$$I_{loop} = MR^2 \Rightarrow 1$$

$$K_i + U_i = K_f + U_f$$

$$0 + Mgh = \frac{1}{2} M V_{cm}^2 + \frac{1}{2} I_{cm} \left(\frac{V_{cm}}{R} \right)^2$$

$$2Mgh = M V_{cm}^2 + \frac{I_{cm} \left(\frac{V_{cm}}{R} \right)^2}{R^2}$$

① Speed of cm only depends on the shape not the mass.

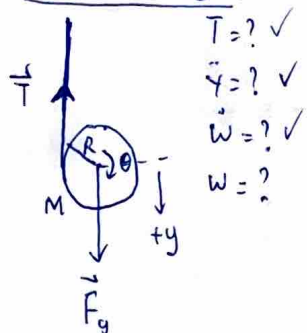
$$2Mgh = \left(M + \frac{I_{cm}}{R^2} \right) V_{cm}^2 \rightarrow V_{cm} = \sqrt{\frac{2Mgh}{M + \frac{I_{cm}}{R^2}}} = \sqrt{\frac{2gh}{1 + \frac{I_{cm}}{MR^2}}}$$

② All objects with the same shape will have the same final velocity.

③ The smaller the ratio I_{cm}/MR^2 the faster the object moves.

④ Any sphere > Any cylinder > Any loop

YOYO PROBLEM:



$$\begin{aligned} T &=? \checkmark \\ \ddot{y} &=? \checkmark \\ \dot{\omega} &=? \checkmark \\ \omega &=? \end{aligned}$$

$$F_g - T = M\ddot{y} \rightarrow Mg - \frac{I\ddot{\theta}}{R} = M\ddot{y} \quad I = \frac{1}{2}MR^2 \text{ for cylinder.}$$

$$N = TR = I\ddot{\theta}$$

$$T = \frac{I\ddot{\theta}}{R}$$

$$Mg - \frac{MR^2}{2} \frac{\ddot{\theta}}{R} = M\ddot{y} \rightarrow \ddot{y} = g - \frac{R\ddot{\theta}}{2}$$

$$y = R\theta \Rightarrow \ddot{y} = g - \frac{R}{2} \ddot{\theta} \quad \frac{3\ddot{y}}{2} = g \Rightarrow \ddot{y} = \frac{2g}{3}$$

$$\begin{aligned} \dot{y} &= R\dot{\theta} \\ \ddot{y} &= R\ddot{\theta} \end{aligned}$$

$$\ddot{\theta} = \dot{\omega} = \frac{\ddot{y}}{R} = \frac{2g}{3R} \Rightarrow T = \frac{1}{2}MR^2 \frac{2g}{3R} = \frac{Mg}{3} = \frac{F_g}{3}$$

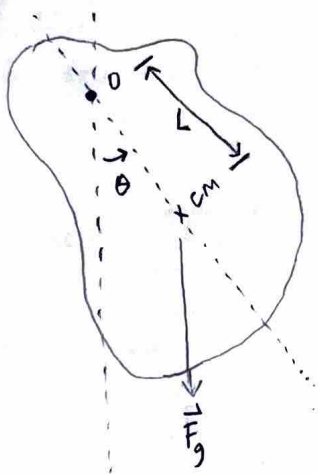
$$\dot{\theta} = \frac{\dot{y}}{R} \quad \int \frac{dy}{dt} = \int \frac{2g}{3} dt \rightarrow \dot{y} = \frac{2g}{3}t + 0 \Rightarrow \dot{\theta} = \frac{2g}{3R}t$$

PHYSICAL PENDULUM

In 11.11.2024 class

- ① Rot. inert. of cylindrical shell
- ② Rot. inert. solid cylinder $\leftrightarrow$ and solid disc.
- ③ Rot. inert. of solid sphere
- ④ Yo-yo problem $\leftrightarrow$ acceleration of cylinder rolling on an inclined plane without slipping
- ⑤ Physical pendulum
- ⑥ Cylinder, sphere, hoop problem which one will come to bottom first.
- ⑦ Sphere pulled by another object going down on an inclined plane

Example:



Physical / Compound pendulum

The body rotates about an axis passing through O. due to the gravitational force acting on the center of mass.

For small oscillations, find the frequency and the period of oscillation if the mass of the body is M and radius of gyration is k.

- **RADIUS OF GYRATION:** Rotational inertia about a given axis of rotation is $I = Mk^2$.

$$N = I\ddot{\theta} \Rightarrow MgL \sin \theta = I\ddot{\theta} \quad \text{small oscillations} \Rightarrow \sin \theta \approx \theta$$

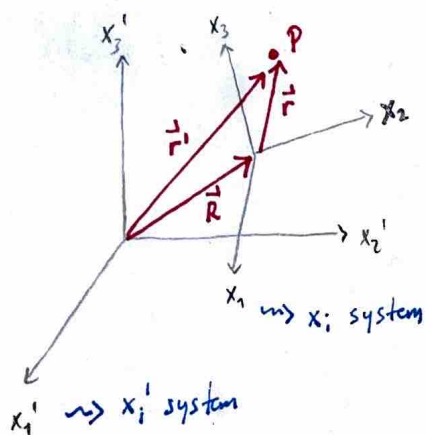
$$\Rightarrow -MgL\theta = I\ddot{\theta} \Rightarrow \ddot{\theta} + \frac{MgL}{I}\theta = 0$$

$$\omega^2 = \frac{MgL}{I}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{MgL}{I}} = \frac{1}{2\pi} \sqrt{\frac{MgL}{Mk^2}} = \frac{1}{2\pi} \sqrt{\frac{gL}{k^2}}$$

$$T = \frac{1}{f} = 2\pi \sqrt{\frac{k^2}{gL}}$$

Rotating coordinate Systems:



x_1', x_2', x_3' : fixed axes inertial reference frame

x_1, x_2, x_3 : axes that in motion w.r.t the other.

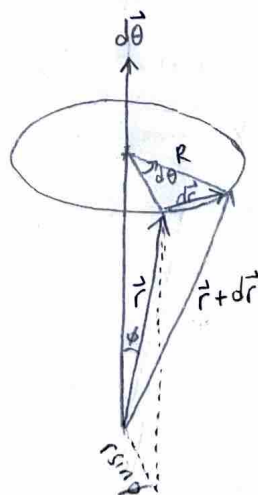
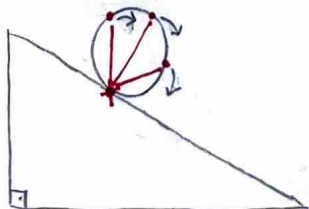
in particular they are "rotating" axes.

For point P:

$$\vec{r}' = \vec{R} + \vec{r}$$

$\vec{r}'$: position vector w.r.t. rotating.
 $\vec{r}$: position vector w.r.t. fixed.
 $\vec{R}$: locates the origin of rotating in the fixed.

We can always represent an infinitesimal displacement by pure rotation about some axis called the instantaneous axis of rotation. For example, the instantaneous motion of a disk rolling down an inclined plane can be described as a rotation about the point of contact between the disk and plane.



$$d\vec{r} = d\vec{\theta} \times \vec{r}$$

$$|d\vec{r}| = d\theta \underbrace{r \sin \phi}_R = d\theta R$$

$$|d\vec{r}| = d\theta R$$

If x_i system undergoes an infinitesimal rotation $d\theta$, corresponding to some arbitrary infinitesimal displacement, the motion of P (and we are assuming here that the P is at rest in x_i system) can be described by

→ this determines the instantaneous axis of rot. of x_i system

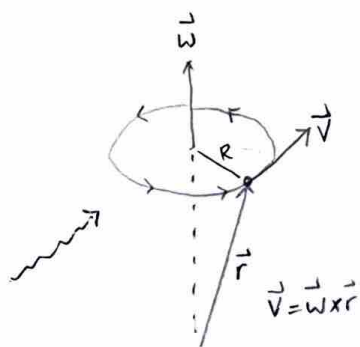
$$(d\vec{r})_{\text{fixed}} = d\vec{\theta} \times \vec{r}$$

The quantity $d\vec{r}$ is measured in x_i system.

★ From the point of view of the rotating system $\vec{r}$ does not change; $\vec{r}$ is fixed in rotating coordinate system.

$$\left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \underbrace{\frac{d\vec{\theta}}{dt}}_{\vec{\omega}} \times \vec{r} \quad (\vec{r} \text{ is fixed during } dt)$$

$$\left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \vec{\omega} \times \vec{r} \quad (\text{for } P \text{ fixed in } x_i \text{ system})$$



If we also allow point P to have a velocity $(d\vec{r}/dt)_{\text{rotating}}$ with respect to the x_i system

$$\left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{r}$$

Example: Consider a vector $\vec{r} = x_1 \hat{i} + x_2 \hat{j} + x_3 \hat{k}$ in the rotating system.

Assume that both systems have the same origin.

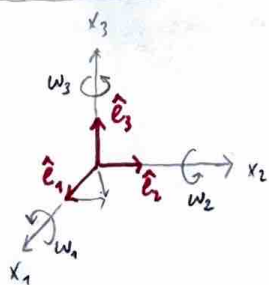
Find $\dot{\vec{r}}$ in the fixed system by direct differentiation if the angular frequency of the rotating system is $\vec{\omega}$ in the fixed system.

$$\left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \frac{d}{dt} \left(\sum_i x_i \hat{e}_i \right) \quad \hat{e}_i \text{ are unit vectors.}$$

$$= \sum_i \dot{x}_i \hat{e}_i + \sum_i x_i \dot{\hat{e}}_i \Rightarrow \sum_i \dot{x}_i \hat{e}_i \text{ : velocity in the rotating system}$$

$$= \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \sum_i x_i \dot{\hat{e}}_i \quad \dot{\hat{e}}_i \text{ : How unit vectors of } x_i \text{ system change with respect to } x_i' \text{ system.}$$

How to calculate $\dot{\hat{e}}_i$?



ω_i : components of the angular velocity; $\vec{\omega} = \omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3$
 $\hat{e}_i$: unit vectors.

ω_2 : turns $\hat{e}_1$ in $-\hat{e}_3$ direction
 ω_3 : turns $\hat{e}_1$ in $\hat{e}_2$ direction

$$\frac{d\hat{e}_1}{dt} = \omega_3 \hat{e}_2 - \omega_2 \hat{e}_3, \quad \frac{d\hat{e}_2}{dt} = -\omega_3 \hat{e}_1 + \omega_1 \hat{e}_3, \quad \frac{d\hat{e}_3}{dt} = \omega_2 \hat{e}_1 - \omega_1 \hat{e}_2$$

In a compact form $\dot{\hat{e}}_i = \vec{\omega} \times \hat{e}_i$

$$\Rightarrow \left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \sum_i \omega_i x_i \hat{e}_i = \vec{v}_r + \vec{\omega} \times \vec{r} \quad \text{the same result as we found above.}$$

$$\left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{r} \quad \text{This equation is not limited to position vector } \vec{r}.$$

It is also valid for an arbitrary vector $\vec{Q}$.

$$\boxed{\left(\frac{d\vec{Q}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{Q}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{Q}} \quad \text{i.e. it can be velocity, acceleration,}$$

- For example, if we consider the angular acceleration $\vec{\omega}$,

$$\left(\frac{d\vec{\omega}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{\omega}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{\omega} = \left(\frac{d\vec{\omega}}{dt}\right)_{\text{rotating}} \rightarrow \text{it is the same in both fixed and rotating systems}$$

Velocity in the fixed coordinate system.

$$\begin{aligned} \text{From } \vec{r}' = \vec{R} + \vec{r} : \quad \left(\frac{d\vec{r}'}{dt}\right)_{\text{fixed}} &= \left(\frac{d\vec{R}}{dt}\right)_{\text{fixed}} + \left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} \\ &= \left(\frac{d\vec{R}}{dt}\right)_{\text{fixed}} + \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{r} \end{aligned}$$

If we define:

$$\left. \begin{aligned} \vec{V}_f &\equiv \dot{\vec{r}}_f \equiv \left(\frac{d\vec{r}'}{dt}\right)_{\text{fixed}} \\ \vec{V} &\equiv \dot{\vec{R}}_f \equiv \left(\frac{d\vec{R}}{dt}\right)_{\text{fixed}} \\ \vec{V}_r &\equiv \dot{\vec{r}}_r \equiv \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} \end{aligned} \right\}$$

$$\boxed{\vec{V}_f = \vec{V} + \vec{V}_r + \vec{\omega} \times \vec{r}}$$

$\vec{V}_f$: velocity relative to fixed axes

$\vec{V}$: linear velocity of the moving origin

$\vec{V}_r$: velocity relative to rotating axes

$\vec{\omega}$: angular velocity of the rotating axes

$\vec{\omega} \times \vec{r}$: velocity due to the rotation of moving axes

Inertia Tensor

- Rigid body composed of n particles of masses m_α ($\alpha = 1, 2, \dots, n$)
- If body rotates with an instantaneous angular velocity $\vec{\omega}$ about some point fixed w.r.t the body coordinate system and if this point moves with linear velocity $\vec{V}$ w.r.t a fixed coordinate system, then the instantaneous velocity of the α th particle in the fixed system

$$\vec{V}_\alpha = \vec{V} + \vec{V}_r + \vec{\omega} \times \vec{r}_\alpha \rightarrow \text{However, we are considering a rigid body}$$

$\Rightarrow \vec{V}_r = 0$. Particles do not move in the rotating axes. Their positions are fixed.

$$\vec{V}_\alpha = \vec{V} + \vec{\omega} \times \vec{r}_\alpha$$

$\rightarrow$ We removed f subscript, we mean that all velocities are measured in the fixed axes.

Kinetic energy of the α th particle:

$$T_\alpha = \frac{1}{2} m_\alpha V_\alpha^2 \quad \xrightarrow{\text{total KE}} \quad T = \frac{1}{2} \sum_\alpha m_\alpha (\vec{V} + \vec{\omega} \times \vec{r}_\alpha)^2$$

$$\Rightarrow T = \frac{1}{2} \sum_\alpha m_\alpha \vec{V}^2 + \sum_\alpha m_\alpha \vec{V} \cdot \vec{\omega} \times \vec{r}_\alpha + \frac{1}{2} \sum_\alpha m_\alpha (\vec{\omega} \times \vec{r}_\alpha)^2$$

- This is a general expression for the KE and it is valid for any choice of the origin from which the vectors $\vec{r}_\alpha$ are measured. But if we make the origin of the body coordinate system coincide with the center of mass of the object, a considerable simplification results.

Consider the middle term in RHS

$$\sum_{\alpha} m_{\alpha} \vec{V} \cdot \vec{\omega} \times \vec{r}_{\alpha} = \vec{V} \cdot \vec{\omega} \times \underbrace{\left(\sum_{\alpha} m_{\alpha} \vec{r}_{\alpha} \right)}_{M\vec{R}}$$

$$\vec{R} = \frac{1}{M} \sum_{\alpha} m_{\alpha} \vec{r}_{\alpha} : \text{COM vector.}$$

If we measure all the distances $\vec{r}_{\alpha}$ from the center of mass (origin of the body coordinate system is the same as center of mass) $M\vec{R}$ vanishes. $\vec{R} \equiv 0$ by definition. Therefore,

$$T = \frac{1}{2} \sum_{\alpha} m_{\alpha} \vec{V}^2 + \frac{1}{2} \sum_{\alpha} m_{\alpha} (\vec{\omega} \times \vec{r}_{\alpha})^2$$

$$T_{\text{trans}} = \frac{1}{2} \sum_{\alpha} m_{\alpha} \vec{V}^2 = \frac{1}{2} M V^2$$

$$T_{\text{rot}} = \frac{1}{2} \sum_{\alpha} m_{\alpha} (\vec{\omega} \times \vec{r}_{\alpha})^2$$

$$T = T_{\text{trans}} + T_{\text{rot}}$$

The rotational K.E can be calculated by considering the following identity:

$$(\vec{A} \times \vec{B})^2 = A^2 B^2 - (\vec{A} \cdot \vec{B})^2$$

$$\Rightarrow T_{\text{rot}} = \frac{1}{2} \sum_{\alpha} m_{\alpha} \left[\omega^2 r_{\alpha}^2 - (\vec{\omega} \cdot \vec{r}_{\alpha})^2 \right] \rightarrow$$

We will express T_{rot} by using the components ω_i and $r_{\alpha,i}$ of the vectors $\vec{\omega}$ and $\vec{r}_{\alpha}$.

$\vec{r}_{\alpha} = (x_{\alpha,1}, x_{\alpha,2}, x_{\alpha,3})$ in the body system, so we will denote $r_{\alpha,i} \equiv x_{\alpha,i}$

$$T_{\text{rot}} = \frac{1}{2} \sum_{\alpha} m_{\alpha} \left[\left(\sum_i \omega_i^2 \right) \left(\sum_k x_{\alpha,k}^2 \right) - \left(\sum_i \omega_i x_{\alpha,i} \right) \left(\sum_j \omega_j x_{\alpha,j} \right) \right]$$

Definition: Kronecker delta

$$\delta_{ij} \equiv \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

We can use $\omega_i = \sum_j \omega_j \delta_{ij} = \omega_1 \delta_{i,1} + \omega_2 \delta_{i,2} + \omega_3 \delta_{i,3}$

$$\sum_i \omega_i^2 = \sum_{i,j} \omega_i \omega_j \delta_{ij} \quad \text{also} \quad \sum_i \omega_i x_{\alpha,i} \sum_j \omega_j x_{\alpha,j} = \sum_{i,j} \omega_i \omega_j x_{\alpha,i} x_{\alpha,j}$$

$$\begin{aligned} \Rightarrow T_{\text{rot}} &= \frac{1}{2} \sum_{\alpha} \sum_{i,j} m_{\alpha} \left[\omega_i \omega_j \delta_{ij} \left(\sum_k x_{\alpha,k}^2 \right) - \omega_i \omega_j x_{\alpha,i} x_{\alpha,j} \right] \\ &= \frac{1}{2} \sum_{i,j} \omega_i \omega_j \sum_{\alpha} m_{\alpha} \left(\delta_{ij} \sum_k x_{\alpha,k}^2 - x_{\alpha,i} x_{\alpha,j} \right) \end{aligned}$$

We will define

$$I_{ij} \equiv \sum_{\alpha} m_{\alpha} \left(\delta_{ij} \sum_k x_{\alpha,k}^2 - x_{\alpha,i} x_{\alpha,j} \right)$$

$$\Rightarrow T_{\text{rot}} = \frac{1}{2} \sum_{i,j} I_{ij} \omega_i \omega_j$$

Remember that we have found out the total energy of a system of particles whose relative distances were fixed

$$T = \frac{1}{2} I \omega^2 \quad I = \sum_{i=1}^N m_i r_i^2$$

T is the most restricted version of T_{rot} .

Properties of I_{ij} or $[\vec{I}]$

- $i, j = 1, 2, 3 \Rightarrow [\vec{I}]$ has 9 elements in total.
- It is a tensor; it has the form of a 3×3 matrix. (It is called "the inertia tensor".)
- It is a proportionality factor between the rotational K.E and the angular velocity.
- It has dimensions $[M][L]^2$.

The elements of $[\vec{I}]$: $I_{ij} = \sum_{\alpha} m_{\alpha} \left(\delta_{ij} \sum_k x_{\alpha,k}^2 - x_{\alpha,i} x_{\alpha,j} \right)$

$$\begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} = [\vec{I}]$$

$$[\vec{I}] = \begin{bmatrix} \sum_{\alpha} m_{\alpha} (x_{\alpha,2}^2 + x_{\alpha,3}^2) & - \sum_{\alpha} m_{\alpha} x_{\alpha,1} x_{\alpha,2} & - \sum_{\alpha} m_{\alpha} x_{\alpha,1} x_{\alpha,3} \\ - \sum_{\alpha} m_{\alpha} x_{\alpha,2} x_{\alpha,1} & \sum_{\alpha} m_{\alpha} (x_{\alpha,1}^2 + x_{\alpha,3}^2) & - \sum_{\alpha} m_{\alpha} x_{\alpha,2} x_{\alpha,3} \\ - \sum_{\alpha} m_{\alpha} x_{\alpha,3} x_{\alpha,1} & - \sum_{\alpha} m_{\alpha} x_{\alpha,3} x_{\alpha,2} & \sum_{\alpha} m_{\alpha} (x_{\alpha,1}^2 + x_{\alpha,2}^2) \end{bmatrix}$$

This was useful when we were dealing with summations.

It is hard to recognize this form of the $[\vec{I}]$. By redefining $(x_{\alpha,1}, x_{\alpha,2}, x_{\alpha,3}) \rightarrow (x_{\alpha}, y_{\alpha}, z_{\alpha})$ and $r_{\alpha}^2 = x_{\alpha}^2 + y_{\alpha}^2 + z_{\alpha}^2$:

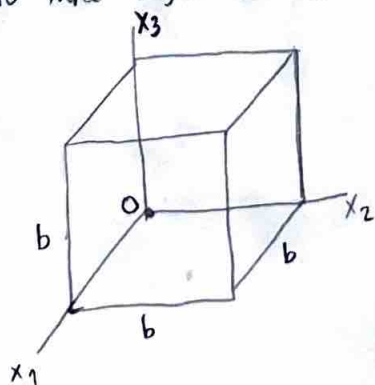
$$[\vec{I}] = \begin{bmatrix} \sum_{\alpha} m_{\alpha} (r_{\alpha}^2 - x_{\alpha}^2) & - \sum_{\alpha} m_{\alpha} x_{\alpha} y_{\alpha} & - \sum_{\alpha} m_{\alpha} x_{\alpha} z_{\alpha} \\ - \sum_{\alpha} m_{\alpha} y_{\alpha} x_{\alpha} & \sum_{\alpha} m_{\alpha} (r_{\alpha}^2 - y_{\alpha}^2) & - \sum_{\alpha} m_{\alpha} y_{\alpha} z_{\alpha} \\ - \sum_{\alpha} m_{\alpha} z_{\alpha} x_{\alpha} & - \sum_{\alpha} m_{\alpha} z_{\alpha} y_{\alpha} & \sum_{\alpha} m_{\alpha} (r_{\alpha}^2 - z_{\alpha}^2) \end{bmatrix}$$

- The diagonal elements I_{11}, I_{22} and I_{33} are called "moments of inertia" about the x_1, x_2 and x_3 axes, respectively and the negatives of off-diagonal elements are called the "products of inertia".
- The inertia tensor is symmetrical: $I_{ij} = I_{ji} \Rightarrow$ There are only six independent elements.
- It is composed of additive elements. For a continuous distribution of matter with mass density $\rho(\vec{r})$:

$$I_{ij} = \int_V \rho(\vec{r}) \left(\delta_{ij} \sum_k x_k^2 - x_i x_j \right) dV$$

where $dV = dx_1 dx_2 dx_3$ element of volume at the position defined by the vector $\vec{r}$. V is the volume of the body.

Example: Calculate the inertia tensor of a homogeneous cube of density ρ , mass M and side of length b . Let one of the corners be the origin and three adjacent edges lie along the coordinate axes.



Origin does not lie at the center of mass. We will return to this later.

$$\begin{aligned} I_{11} &= \rho \int_0^b \int_0^b \int_0^b (x_2^2 + x_3^2) dx_1 dx_2 dx_3 \\ &= \rho b^2 \left[\frac{x_2^3}{3} \right]_0^b + \rho b^2 \left[\frac{x_3^3}{3} \right]_0^b = \rho \frac{b^5}{3} + \rho \frac{b^5}{3} = \frac{2}{3} \rho b^5 \\ &= \frac{2}{3} (\rho b^3) b^2 = \frac{2}{3} M b^2 = I_{22} = I_{33} \end{aligned}$$

$$I_{12} = - \int_0^b \int_0^b \int_0^b x_1 x_2 dx_1 dx_2 dx_3 = - \int_0^b x_1 dx_1 \int_0^b x_2 dx_2 \int_0^b dx_3$$

$$= - \int_0^b \frac{b^2}{2} \frac{b^2}{2} b = - \int_0^b \frac{b^5}{4} = - \frac{1}{4} M b^2 = I_{13} = I_{23}$$

$$[I] = \begin{bmatrix} 2P/3 & -P/4 & -P/4 \\ -P/4 & 2P/3 & -P/4 \\ -P/4 & -P/4 & 2P/3 \end{bmatrix}$$

Define $Mb^2 = \beta$: $I_{11} = I_{22} = I_{33} = 2\beta/3$

$$I_{12} = I_{13} = I_{23} = -\beta/4$$

Angular Momentum:

$$L = \sum_a \vec{r}_a \times \vec{p}_a$$

angular momentum of a body (or system of particles) with respect to some point O fixed in the body coordinate system.
(wrt to fixed coordinates)

How to choose O : It depends on the particular problem. Two choices are important:

(i) If one or more points of the body are fixed (in the fixed coordinate system), O is chosen to coincide with such point.

(ii) If no point is fixed, O is chosen to be the center of mass.

Relative to the body coordinate system (or fixed?) the linear momentum of ath particle.

$$\vec{p}_a = m_a \vec{v}_a = m_a \vec{\omega} \times \vec{r}_a$$

So, the angular momentum of the body becomes,

$$\vec{L} = \sum_a m_a \vec{r}_a \times (\vec{\omega} \times \vec{r}_a)$$

We will make use of the following vector identity

$$\vec{A} \times (\vec{B} \times \vec{A}) = A^2 \vec{B} - \vec{A}(\vec{A} \cdot \vec{B})$$

to express $\vec{L}$:

$$\vec{L} = \sum_a m_a \left[r_a^2 \vec{\omega} - \vec{r}_a (\vec{r}_a \cdot \vec{\omega}) \right]$$

$$\vec{r}_a = \begin{pmatrix} x_{a,1} \\ x_{a,2} \\ x_{a,3} \end{pmatrix} \quad \vec{\omega} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

r_a^2 : a number
 $\vec{r}_a \cdot \vec{\omega}$: a number

$\vec{\omega}$: vector
 $\vec{r}_a$: vector $\rightarrow \left\{ [A \text{ number}] \vec{\omega} - [Another \text{ number}] \vec{r}_a \right\}_i = [L]_i \vec{\omega}_i - [r]_i \vec{\omega}_i$

$$\Rightarrow L_i = \sum_a m_a \left[\left(\sum_k x_{a,k}^2 \right) \omega_i - x_{a,i} \left(\sum_j x_{a,j} \omega_j \right) \right] \rightarrow \omega_i = \sum_j \omega_j \delta_{ij}$$

$$= \sum_a m_a \sum_j \left(\omega_j \delta_{ij} \sum_k x_{a,k}^2 - \omega_j x_{a,i} x_{a,j} \right)$$

$$= \sum_j \omega_j \sum_a m_a \left(\delta_{ij} \sum_k x_{a,k}^2 - x_{a,i} x_{a,j} \right)$$

I_{ij}

$$= \sum_j \omega_j I_{ij} \Rightarrow \boxed{L_i = \sum_j I_{ij} \omega_j} \Rightarrow \boxed{\vec{L} = [\vec{I}] \cdot \vec{\omega}}$$

What does $\vec{L} = [\vec{I}]\vec{\omega}$ mathematically mean?

$[\vec{I}]$: 3x3 matrix
 $\vec{\omega}$: 3x1 vector

$\left. \begin{array}{l} \\ \end{array} \right\} [\vec{I}]\vec{\omega}$ represents matrix / vector multiplication

$$\begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} I_{11}\omega_1 + I_{12}\omega_2 + I_{13}\omega_3 \\ I_{21}\omega_1 + I_{22}\omega_2 + I_{23}\omega_3 \\ I_{31}\omega_1 + I_{32}\omega_2 + I_{33}\omega_3 \end{bmatrix} = \begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix}$$

Components of angular velocity Components of angular momentum

$$L_i = \sum_j I_{ij} \omega_j \text{ as we have found before.}$$

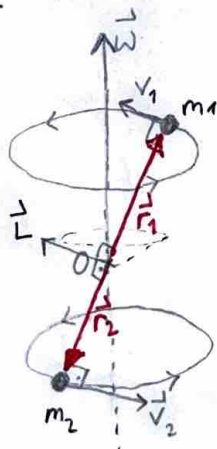
! This result may seem a bit unexpected:

Assume $\vec{\omega} = [\omega_1, 0, 0]$: only in x-direction. If $I_{ij} \neq 0$ for $i \neq j$

then $\vec{L} = [L_1, L_2, L_3]$ with $L_1 \neq 0, L_2 \neq 0, L_3 \neq 0$

★ $\vec{L}$ does not in general have the same direction as the angular velocity vector
 (We will return to this point soon.)

Example: How $\vec{\omega}$ and $\vec{L}$ can be colinear? (Rotating dumbbell)



Assume that stuff connecting m_1 and m_2 is weightless.

$$\vec{V}_a = \vec{\omega} \times \vec{r}_a$$

$$\vec{L} = \sum_a m_a \vec{r}_a \times \vec{V}_a$$

- $\vec{\omega}$ must be directed along the axis of rotation.
- $\vec{L}$ must be perpendicular to the line connecting m_1 and m_2 .

★ Note that $\vec{L}$ does not remain constant in time but rotates with an angular velocity ω in such a way that it traces out a cone whose axis is the axis of rotation.
 Therefore $\dot{\vec{L}} \neq 0$. We also know that $\dot{\vec{L}} = \vec{\tau}$.

⇒ Keep in mind that we must constantly apply a torque to keep the dumbbell rotating as above.

Rotational Kinetic Energy in terms of $\vec{\omega}$ and $\vec{L}$

$$L_i = \sum_j I_{ij} \omega_j \xrightarrow[\text{sides with } \frac{1}{2} \omega_i]{\text{mult. both}} \frac{1}{2} \omega_i L_i = \frac{1}{2} \omega_i \sum_j I_{ij} \omega_j \xrightarrow[\text{sum over } i]{\text{sum over } i} \sum_i \frac{1}{2} \omega_i L_i = \frac{1}{2} \sum_{ij} I_{ij} \omega_i \omega_j = T_{\text{rot}}$$

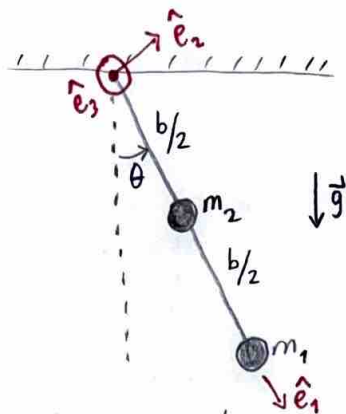
$$\Rightarrow T_{\text{rot}} = \frac{1}{2} \sum_i \omega_i L_i = \frac{1}{2} \vec{\omega} \cdot \vec{L} \rightsquigarrow T_{\text{rot}} = \frac{1}{2} \vec{\omega} \cdot \vec{L} \rightsquigarrow T_{\text{rot}} = \frac{1}{2} \vec{\omega} \cdot [\underbrace{[\vec{I}]}_{\text{VECTOR}}] \cdot \underbrace{\vec{\omega}}_{\text{SCALAR}}$$

★ KEEP IN MIND: [VECTOR] · [VECTOR] = [SCALAR]

[MATRIX] · [VECTOR] = [VECTOR]

COROLLARY: [VECTOR] · [MATRIX] · [VECTOR] = [SCALAR]

Example:



m_1 and m_2 fixed on a rigid rod of length b .

- Calculate the inertia tensor $[\vec{I}]$.
- Calculate the angular momentum vector $\vec{L}$.
- Write down the Newton's 2nd law for rotations.
- Find the frequency of small oscillations if the pendulum swings in a plane.

Let fixed and body systems have their origin at the pendulum pivot point.

$$\vec{\omega} = \omega_3 \hat{e}_3 = \dot{\theta} \hat{e}_3 \quad I_{ij} \equiv \sum_{\alpha} m_{\alpha} \left(\delta_{ij} \sum_k x_{\alpha,k}^2 - x_{\alpha,i} x_{\alpha,j} \right)$$

All the mass is along $\hat{e}_1$ with $x_{1,1} = b$ and $x_{2,1} = b/2$ all other components of $x_{\alpha,k}$ equal 0.

$$I_{ij} = m_1 \left(\delta_{ij} \sum_k x_{1,k}^2 - x_{1,i} x_{1,j} \right) + m_2 \left(\delta_{ij} \sum_k x_{2,k}^2 - x_{2,i} x_{2,j} \right) \quad (k=1 \rightarrow \text{No } y \text{ or } z \text{ coordinates})$$

$$= m_1 \left(\delta_{ij} x_{1,1}^2 - x_{1,i} x_{1,j} \right) + m_2 \left(\delta_{ij} x_{2,1}^2 - x_{2,i} x_{2,j} \right)$$

$$I_{11} = 0 \quad I_{22} = m_1 (x_{1,1}^2 - x_{1,2}^2) + m_2 (x_{2,1}^2 - 0) = m_1 b^2 + m_2 \frac{b^2}{4}$$

$$I_{33} = m_1 (x_{1,1}^2 - x_{1,3}^2) + m_2 (x_{2,1}^2 - x_{2,3}^2) = m_1 b^2 + m_2 \frac{b^2}{4}$$

For off-diagonal elements:

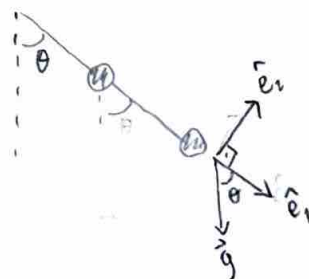
$$I_{ij} = -m_1 x_{1,i} x_{1,j} - m_2 x_{2,i} x_{2,j} = 0 \quad i \neq j$$

$$[\vec{I}] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & m_1 b^2 + m_2 \frac{b^2}{4} & 0 \\ 0 & 0 & m_1 b^2 + m_2 \frac{b^2}{4} \end{bmatrix}$$

$$L_i = \sum_j I_{ij} \omega_j \quad \text{or} \quad \vec{L} = [\vec{I}] \vec{\omega}$$

$$\vec{L} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & m_1 b^2 + m_2 \frac{b^2}{4} & 0 \\ 0 & 0 & m_1 b^2 + m_2 \frac{b^2}{4} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ (m_1 b^2 + m_2 \frac{b^2}{4}) \dot{\theta} \end{bmatrix}$$

The only external force acting on the system is gravity, which causes torque on the system.



$$\vec{N} = \dot{\vec{L}} \rightarrow (m_1 b^2 + m_2 \frac{b^2}{4}) \ddot{\theta} \hat{e}_3 = \sum_{\alpha} \vec{r}_{\alpha} \times \vec{F}_{\alpha}$$

$$\vec{g} = g \cos \theta \hat{e}_1 + g \sin \theta \hat{e}_2$$

$$\vec{r}_1 \times \vec{F}_1 = b \hat{e}_1 \times (\cos \theta \hat{e}_1 - \sin \theta \hat{e}_2) m_1 g = -m_1 g b \sin \theta \hat{e}_3$$

$$\vec{r}_2 \times \vec{F}_2 = \frac{b}{2} \hat{e}_1 \times (\cos \theta \hat{e}_1 - \sin \theta \hat{e}_2) m_2 g = -m_2 g \frac{b}{2} \sin \theta \hat{e}_3$$

$$(m_1 b^2 + m_2 \frac{b^2}{4}) \ddot{\theta} \hat{e}_3 = -\hat{e}_3 (m_1 g b \sin \theta + m_2 g \frac{b}{2} \sin \theta)$$

$$b \left(m_1 + \frac{m_2}{4} \right) \ddot{\theta} = -g \sin \theta \left(m_1 + \frac{m_2}{2} \right) \quad \sin \theta \approx \theta \Rightarrow \ddot{\theta} + \frac{g \left(m_1 + \frac{m_2}{2} \right)}{b \left(m_1 + \frac{m_2}{4} \right)} \theta = 0 \Rightarrow \omega_0^2 = \frac{g}{b} \frac{m_1 + \frac{m_2}{2}}{m_1 + \frac{m_2}{4}}$$

For $m_1 \gg m_2$: $\omega_0^2 \approx \frac{g}{b}$ For $m_1 \ll m_2$: $\omega_0^2 \approx \frac{2}{3} \frac{g}{b}$ as we expected

Principal axes of inertia:

Can we find some axes that the body rotate about ^{both} axes so that the angular momentum and angular velocity will be directed in the same direction of this axes?
We call such axes **PRINCIPAL AXES OF INERTIA**.

$$[\vec{I}] \vec{\omega} = I \vec{\omega} \quad \{[\vec{I}] - I \mathbb{I}\} \vec{\omega} = \vec{0} \quad \text{Eigenvalue problem} \rightarrow \text{which } I_x \text{ and } \vec{\omega}'s \text{ satisfy this equation}$$

λ is a scalar, and it is rotational inertia.

Then

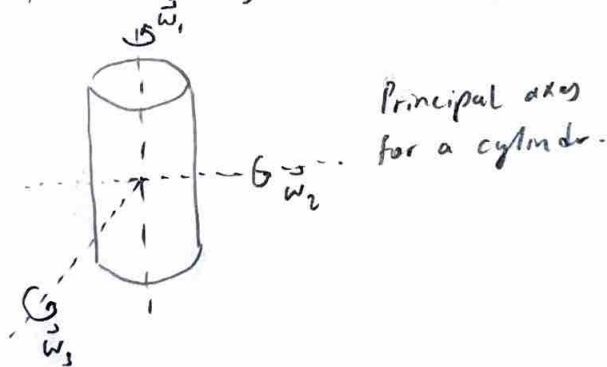
$$\begin{vmatrix} I_{11} - I & I_{12} & I_{13} \\ I_{21} & I_{22} - I & I_{23} \\ I_{31} & I_{32} & I_{33} - I \end{vmatrix} = 0 \quad \text{To find } I_x's \text{ and } \vec{\omega}'s \text{ determinant must vanish.}$$

★ We will see that this determinant will yield a cubic equation. Therefore, there will be three roots of this equation; I_1, I_2, I_3 . These values are called **PRINCIPAL MOMENTS OF INERTIA**. And there will be three axes $\vec{\omega}_1, \vec{\omega}_2, \vec{\omega}_3$ corresponding to the values I_1, I_2 and I_3 .

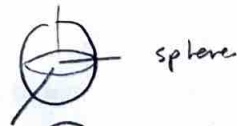
★ If a body rotates about, say $\vec{\omega}_1$, with rotational inertia I_1

$$[\vec{I}] \vec{\omega}_1 = I_1 \vec{\omega}_1 = \vec{L} \rightarrow \vec{L} \text{ and } \vec{\omega}_1 \text{ have the same direction.}$$

★ In most of the problems encountered in rigid-body dynamics, the bodies are of some regular shape, so we can determine the principal axes merely by examining the symmetry of the body. e.g. cylinder.



★ If a body has $I_1 = I_2 = I_3$: spherical top e.g.



If $I_1 = I_2 \neq I_3$: symmetrical top e.g.



If $I_1 \neq I_2 \neq I_3$: asymmetrical top.

Example: Find the principal moments of inertia and principle axes of the cube we considered in the previous problem. (rotating around its corner)

We have found: $\rightarrow$ We choose one of the corners of the cube as the origin and obviously we didn't take one of the principal axes.

$$[\vec{I}] = \begin{bmatrix} \frac{2}{3} \beta & -\frac{1}{4} \beta & -\frac{1}{4} \beta \\ -\frac{1}{4} \beta & \frac{2}{3} \beta & -\frac{1}{4} \beta \\ -\frac{1}{4} \beta & -\frac{1}{4} \beta & \frac{2}{3} \beta \end{bmatrix}$$

For example if the cube rotates about z axis then $\vec{\omega} = \omega_3 \hat{e}_3$
and

$$\left. \begin{aligned} L_1 &= -\frac{1}{4} \beta \omega_3 \\ L_2 &= -\frac{1}{4} \beta \omega_3 \\ L_3 &= \frac{2}{3} \beta \omega_3 \end{aligned} \right\} \rightarrow \vec{L} = M b^2 \omega_3 \left(-\frac{1}{4} \hat{e}_1 - \frac{1}{4} \hat{e}_2 + \frac{2}{3} \hat{e}_3 \right)$$

which is not in the same direction as $\vec{\omega}$.

To find principal axes and principal moments of inertia

$$\begin{vmatrix} \frac{2}{3}\beta - I & -\frac{1}{4}\beta & -\frac{1}{4}\beta \\ -\frac{1}{4}\beta & \frac{2}{3}\beta - I & -\frac{1}{4}\beta \\ -\frac{1}{4}\beta & -\frac{1}{4}\beta & \frac{2}{3}\beta - I \end{vmatrix} = \begin{vmatrix} \frac{2}{3}\beta - I & -\frac{1}{4}\beta & -\frac{1}{4}\beta \\ -\frac{11}{12}\beta + I & \frac{11}{12}\beta - I & 0 \\ -\frac{1}{4}\beta & -\frac{1}{4}\beta & \frac{2}{3}\beta - I \end{vmatrix} = \left(\frac{11}{12}\beta - I\right) \begin{vmatrix} \frac{2}{3}\beta - I & -\frac{1}{4}\beta & -\frac{1}{4}\beta \\ -1 & 1 & 0 \\ -\frac{1}{4}\beta & -\frac{1}{4}\beta & \frac{2}{3}\beta - I \end{vmatrix} = 0$$

$$\Rightarrow \left(\frac{11}{12}\beta - I\right) \left\{ \left(\frac{2}{3}\beta - I\right)^2 - \frac{1}{4}\beta \left(\frac{2}{3}\beta - I\right) + \left(-\frac{1}{4}\beta\right) \frac{1}{2}\beta \right\} = 0$$

$$\left(\frac{11}{12}\beta - I\right) \left\{ \left(\frac{2}{3}\beta - I\right)^2 - \frac{1}{4}\beta \left(\frac{2}{3}\beta - I\right) - \frac{1}{8}\beta^2 \right\} = 0$$

$$\left(\frac{4}{9}\beta^2\right) - \frac{4}{3}\beta I + I^2 - \frac{2}{12}\beta^2 + \frac{1}{4}\beta I - \frac{1}{8}\beta^2$$

$$\left(\frac{4}{9} - \frac{2}{12} - \frac{1}{8}\right)\beta^2 + \beta I \left(\frac{1}{4} - \frac{4}{3}\right) + I^2$$

$$\frac{32 - 12 - 9}{72}\beta^2 - \frac{13}{12}\beta I + I^2 = \frac{11}{72}\beta^2 - \frac{13}{12}\beta I + I^2 = \left(\frac{11}{12}\beta - I\right) \left(\frac{1}{6}\beta - I\right)$$

$$\Rightarrow \left(\frac{11}{12}\beta - I\right) \left(\frac{11}{12}\beta - I\right) \left(\frac{1}{6}\beta - I\right) = 0 \Rightarrow I_1 = \frac{11}{12}\beta, I_2 = \frac{11}{12}\beta, I_3 = \frac{1}{6}\beta$$

$$[\vec{I}] = \begin{bmatrix} \frac{1}{6}\beta & 0 & 0 \\ 0 & \frac{11}{12}\beta & 0 \\ 0 & 0 & \frac{11}{12}\beta \end{bmatrix}$$

Why diagonal? See the discussion at the back of pg. 63

The roots I_2 and I_3 are identical. So I_1 must correspond to axis of symmetry. Let's calculate the principle axis ass. with $I_1 = \frac{1}{6}\beta$

$$\frac{2}{3}\beta - \frac{1}{6}\beta = \frac{12\beta}{18} - \frac{3\beta}{18} = \frac{9\beta}{18} = \frac{\beta}{2}$$

$$([\vec{I}] - I_1 \mathbf{1}) \vec{w} = \vec{0}$$

$$\frac{\beta}{2} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

corresponds to

$$w_1 - \frac{w_2}{2} - \frac{w_3}{2} = 0$$

$$-\frac{1}{2}w_1 + w_2 - \frac{1}{2}w_3 = 0$$

$$-\frac{1}{2}w_1 - \frac{1}{2}w_2 + w_3 = 0$$

$$w = w_2 = w_3 : \text{from second row}$$

$$w_1 - \frac{w_2}{2} - \frac{w_3}{2} = 0 \rightarrow w_1 = w$$

$$\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{3}{2} & -\frac{3}{2} \\ 0 & -\frac{1}{2} & \frac{3}{2} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{3}{2} & -\frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} w \\ w \\ w \end{bmatrix}$$

Projections of $\vec{w}$ on the 3 coordinates are all equal.

Principle axis $\rightarrow$ Diagonal! (61)

Moments of inertia for different body coordinate system.

Choosing the origin

as the C.O.M. of the
in body coordinate axes

→ we are capable to write the K.E in terms of
only the rotational and translational parts only.

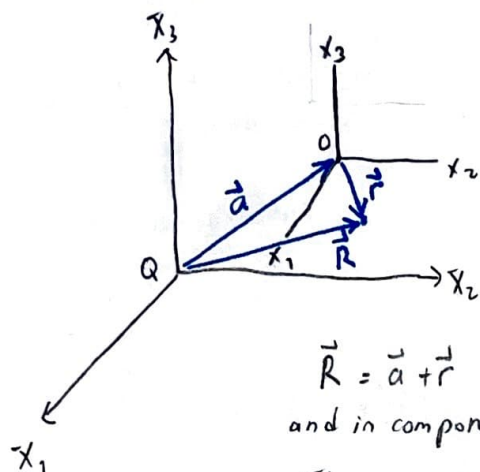
★ However sometimes for certain geometrical objects it may not be convenient to compute the $[I]$ by choosing C.O.M. as the origin.

★ We will consider another coordinates X_i

• Again fixed with respect to the body.

• Yet, the origins of x_i and X_i do not overlap.

• Has the same orientation with x_i .



$$\vec{R} = \vec{a} + \vec{r}$$

and in component form

$$X_i = x_i + a_i$$

Let us assume that the elements of $[I]$ relative to X_i can be written as

$$J_{ij} = \sum_{\alpha} m_{\alpha} \left(\delta_{ij} \sum_k X_{\alpha,k}^2 - X_{\alpha,i} X_{\alpha,j} \right)$$

$$J_{ij} = \sum_{\alpha} m_{\alpha} \left[\delta_{ij} \sum_k (x_{\alpha,k} + a_k)^2 - (x_{\alpha,i} + a_i)(x_{\alpha,j} + a_j) \right]$$

$$= \sum_{\alpha} m_{\alpha} \left[\delta_{ij} \sum_k (x_{\alpha,k}^2 + 2x_{\alpha,k}a_k + a_k^2) - (x_{\alpha,i}x_{\alpha,j} + x_{\alpha,i}a_j + x_{\alpha,j}a_i + a_i a_j) \right]$$

$$= \sum_{\alpha} m_{\alpha} \left(\underbrace{\delta_{ij} \sum_k x_{\alpha,k}^2 - x_{\alpha,i} x_{\alpha,j}}_{I_{ij}} + \sum_k (2x_{\alpha,k}a_k + a_k^2) - (a_i x_{\alpha,j} + a_j x_{\alpha,i} + a_i a_j) \right)$$

$$= I_{ij} + \sum_{\alpha} m_{\alpha} \left(\delta_{ij} \sum_k a_k^2 - a_i a_j \right) + \sum_{\alpha} m_{\alpha} \left(2\delta_{ij} \sum_k x_{\alpha,k} a_k - a_i x_{\alpha,j} - a_j x_{\alpha,i} \right)$$

Each term here involves $\sum_{\alpha} m_{\alpha} x_{\alpha,k}$

$$\sum_{\alpha} m_{\alpha} \left(2\delta_{ij} \sum_k x_{\alpha,k} a_k - a_i x_{\alpha,j} - a_j x_{\alpha,i} \right)$$

→ we know that in x_i , O is located at the C.O.M. $\Rightarrow \sum_{\alpha} m_{\alpha} \vec{r} = 0$

$$= 2\delta_{ij} \sum_k \underbrace{\sum_{\alpha} m_{\alpha} x_{\alpha,k} a_k}_0 - a_i \underbrace{\sum_{\alpha} m_{\alpha} x_{\alpha,j}}_0 - a_j \underbrace{\sum_{\alpha} m_{\alpha} x_{\alpha,i}}_0$$

(each term in the summation) 0

or for the kth component $\sum_{\alpha} m_{\alpha} x_{\alpha,k} = 0$

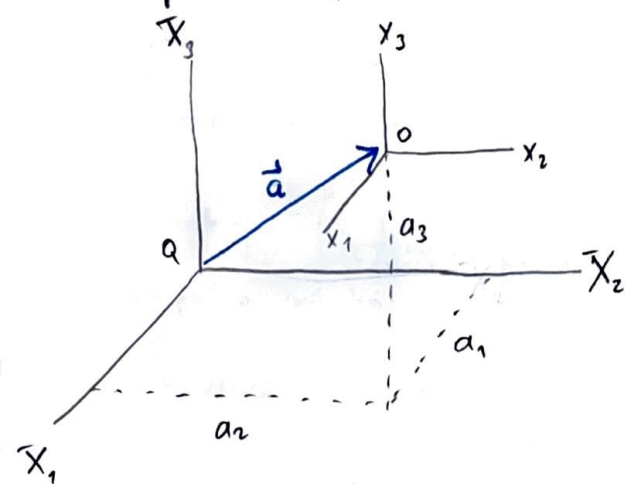
$$\Rightarrow J_{ij} = I_{ij} + \sum_{\alpha} m_{\alpha} \left(\delta_{ij} \sum_k a_k^2 - a_i a_j \right) \quad \text{we also know that } \sum_{\alpha} m_{\alpha} = M \text{ and } \sum_k a_k^2 = a^2$$

$$\Rightarrow \boxed{I_{ij} = J_{ij} - M \left(\delta_{ij} a^2 - a_i a_j \right)}$$

This equation allows us to calculation of elements I_{ij} of the inertia tensor (with origin at the C.O.M.) once those with respect to the X_i -axes are known.

→ STEINER'S PARALLEL
AXES THEOREM.

For example consider,



$$I_{11} = J_{11} + M[a_1^2 + a_2^2 + a_3^2 - a_1^2]$$

$$= J_{11} + M(a_2^2 + a_3^2)$$

Example: Find the inertia tensor of the cube we considered before in a coordinate system with origin at the center of mass.

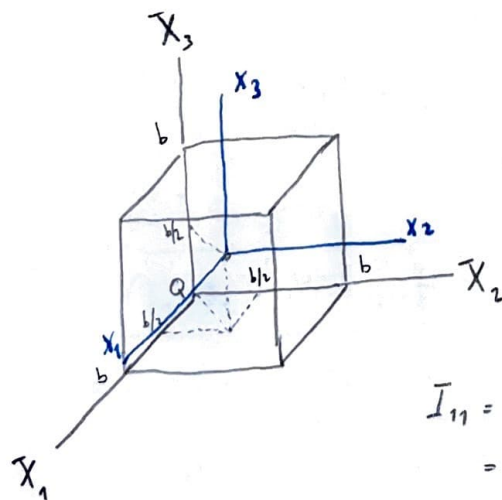
We found that

$$[\vec{J}] = \begin{bmatrix} \frac{2}{3}\beta & -\frac{1}{4}\beta & -\frac{1}{4}\beta \\ -\frac{1}{4}\beta & \frac{2}{3}\beta & -\frac{1}{4}\beta \\ -\frac{1}{4}\beta & -\frac{1}{4}\beta & \frac{2}{3}\beta \end{bmatrix} \quad \text{with } \beta = Mb^2$$

X_i : origin at the corner

X_i : origin at the C.O.M.

C.O.M of the cube $(b/2, b/2, b/2) = \vec{a}$



$$I_{ij} = J_{ij} - M(\delta_{ij} a^2 - a_i a_j)$$

$$J_{11} = J_{22} = J_{33} = \frac{2}{3}\beta$$

$$J_{ij} = -\frac{1}{4}\beta \quad i \neq j$$

$$I_{11} = J_{11} - M(a^2 - a_1^2)$$

$$= J_{11} - M(a_2^2 + a_3^2) = \frac{2}{3}\beta - M\left(\frac{b^2}{4} + \frac{b^2}{4}\right) = \frac{2}{3}\beta - M\frac{b^2}{2}$$

$$= \frac{2}{3}\beta - \frac{1}{3}\beta = \frac{\beta}{3}$$

$$I_{22} = J_{22} - M(a_1^2 + a_3^2) = \beta/6 = I_{33}$$

$$I_{12} = J_{12} + M a_1 a_2 = -\frac{1}{4}\beta + M\frac{b^2}{4} = 0 = I_{13} = I_{23}$$

$$[\vec{I}] = \begin{bmatrix} \beta/6 & 0 & 0 \\ 0 & \beta/6 & 0 \\ 0 & 0 & \beta/6 \end{bmatrix} = \frac{\beta}{6} \mathbb{1}_{3 \times 3}$$

★ If we choose the origin as the center of mass of the cube, principal axes are perpendicular to the faces of the cube. (The eigenvectors of matrix $[\vec{I}]$ are directed along x_1, x_2 and x_3)

→ Principal moments of inertia are all the same

Euler's equations for a force field

$$\left(\frac{d\vec{L}}{dt}\right)_{\text{fixed}} = \vec{N}$$

Here, we explicitly state that the time rate of change of angular momentum is measured from the fixed (inertial) coordinates since Newton's second law is ~~app~~ only applicable in inertial coordinates.

Remember : $\left(\frac{d\vec{Q}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{Q}}{dt}\right)_{\text{body}} + \vec{\omega} \times \vec{Q}$ for an arbitrary vector $\vec{Q}$.
Let's choose $\vec{Q} = \vec{L}$

$$\left(\frac{d\vec{L}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{L}}{dt}\right)_{\text{body}} + \vec{\omega} \times \vec{L}$$

$$\Rightarrow \left(\frac{d\vec{L}}{dt}\right)_{\text{body}} + \vec{\omega} \times \vec{L} = \vec{N}$$

Component of this equation along X_3 axis (body coordinates) :

$$\dot{L}_3 + \omega_1 L_2 - \omega_2 L_1 = N_3$$

We know that,

$$L_i = \sum_j I_{ij} \omega_j \rightarrow \text{But we will choose the principal axes as the body coordinates so the inertia tensor becomes diagonal with such a choice}$$

$$\Rightarrow L_i = \sum_j I_{ij} \delta_{ij} \omega_j$$

$$L_i = I_i \omega_i$$

$$I_{ij} = I_i \delta_{ij}$$

$$\Rightarrow \dot{L}_3 = I_3 \dot{\omega}_3, \quad L_1 = I_1 \omega_1, \quad L_2 = I_2 \omega_2$$

$$\Rightarrow I_3 \dot{\omega}_3 + \omega_1 I_2 \omega_2 - \omega_2 I_1 \omega_1 = N_3$$

$$I_3 \dot{\omega}_3 - (I_1 - I_2) \omega_1 \omega_2 = N_3$$

Similarly the other components can be written as.

$$I_1 \dot{\omega}_1 - (I_2 - I_3) \omega_2 \omega_3 = N_1$$

$$I_2 \dot{\omega}_2 - (I_3 - I_1) \omega_3 \omega_1 = N_2$$

$$I_3 \dot{\omega}_3 - (I_1 - I_2) \omega_1 \omega_2 = N_3$$

→ Euler equations for the motion of a rigid body in a force field.



$$\vec{\omega} = \omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}$$

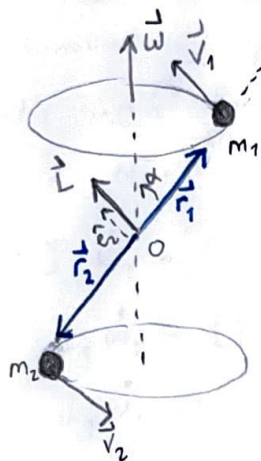
$$\vec{L} = L_1 \hat{i} + L_2 \hat{j} + L_3 \hat{k}$$

$$\vec{\omega} \times \vec{L} = \omega_1 L_2 \hat{i} \times \hat{j} + \omega_1 L_3 \hat{i} \times \hat{k} + \omega_2 L_1 \hat{j} \times \hat{i} + \omega_2 L_3 \hat{j} \times \hat{k} + \omega_3 L_1 \hat{k} \times \hat{i} + \omega_3 L_2 \hat{k} \times \hat{j}$$

$$= \omega_1 L_2 \hat{k} - \omega_1 L_3 \hat{j} - \omega_2 L_1 \hat{k} + \omega_2 L_3 \hat{i} - \omega_3 L_1 \hat{j} + \omega_3 L_2 \hat{i}$$

$$= (\omega_1 L_2 - \omega_2 L_1) \hat{i} + (\omega_1 L_3 - \omega_3 L_1) \hat{j} + (\omega_2 L_3 - \omega_3 L_2) \hat{k}$$

Example: Consider the previous dumbbell problem. Find the angular momentum of the system and the torque required to maintain the motion.



Let $|\vec{r}_1| = |\vec{r}_2| = b$ origin be O, symmetry axis x_3 be along towards m_1 .

$$\vec{L} = \sum_{\alpha} m_{\alpha} \vec{r}_{\alpha} \times \vec{V}_{\alpha}$$

We know that $\vec{L}$ is perpendicular to the shaft and rotates around $\vec{\omega}$ (as the shaft rotates) we can simply choose $\hat{e}_2$ to be along $\vec{L}$.

$$\vec{L} = L_2 \hat{e}_2$$

- If the angle between $\vec{\omega}$ and the shaft is α , the components of $\vec{\omega}$ are

$$\omega_1 = 0$$

$$\omega_2 = \omega \sin \alpha$$

$$\omega_3 = \omega \cos \alpha$$

- Since we have chosen the principle axes as our basis, $[\vec{I}]$ will be diagonalized. So, it is enough to calculate only the principle moments of inertia, I_1, I_2 and I_3 .

$$I_1 = \sum_{\alpha} m_{\alpha} (x_{\alpha,2}^2 + x_{\alpha,3}^2) = m_1 b^2 + m_2 b^2 = (m_1 + m_2) b^2$$

$$I_2 = \sum_{\alpha} m_{\alpha} (x_{\alpha,1}^2 + x_{\alpha,3}^2) = m_1 b^2 + m_2 b^2 = (m_1 + m_2) b^2$$

$$I_3 = \sum_{\alpha} m_{\alpha} (x_{\alpha,1}^2 + x_{\alpha,2}^2) = 0$$

If $[\vec{I}]$ is diagonalized $\vec{L} = [\vec{I}] \cdot \vec{\omega} \rightarrow L_i = I_i \omega_i$

$$L_1 = I_1 \omega_1$$

$$L_2 = I_2 \omega_2 = (m_1 + m_2) b^2 \omega \sin \alpha$$

$$L_3 = I_3 \omega_3 = 0$$

Using Euler's equations:

$$N_1 = (m_1 + m_2) b^2 \dot{\omega}_1 - [(m_1 + m_2) b^2 - 0] \omega^2 \sin \alpha \cos \alpha = - (m_1 + m_2) b^2 \omega^2 \sin \alpha \cos \alpha$$

$$N_2 = (m_1 + m_2) b^2 \dot{\omega}_2 \sin^2 \alpha + (m_1 + m_2) b^2 \omega^2 \sin \alpha \cos \alpha = 0$$

$$N_3 = 0 - (0) = 0$$

* The torque required to maintain the motion if $\dot{\omega} = 0$ is directed along $\hat{e}_1$ axis.

Force-free motion of a symmetric top

- A symmetric top : A rigid body with $I_1 = I_2 \neq I_3$.
- Let's write down Euler's equations by substituting I_1 for I_2 and $N_1 = N_2 = N_3 = 0$

$$\begin{aligned} I_1 \dot{\omega}_1 - (I_1 - I_3) \omega_2 \omega_3 &= 0 \\ I_1 \dot{\omega}_2 - (I_3 - I_1) \omega_3 \omega_1 &= 0 \\ I_3 \dot{\omega}_3 &= 0 \end{aligned} \quad \rightarrow \quad \boxed{\begin{aligned} (I_1 - I_3) \omega_2 \omega_3 - I_1 \dot{\omega}_1 &= 0 \\ (I_3 - I_1) \omega_3 \omega_1 - I_1 \dot{\omega}_2 &= 0 \\ I_3 \dot{\omega}_3 &= 0 \end{aligned}} \quad (*)$$

- The motion is force-free $\rightarrow$ COM of the body is either at rest or in uniform motion w.r.t the fixed frame of reference.

$\Rightarrow$ we will use, without losing any generality, body's COM is at rest and located at the origin of the fixed coordinate system.

★ We consider the case in which $\vec{\omega}$ does not lie along a principle axis of the body. Otherwise, the motion would be trivial.

$\rightarrow$ The first result from (*) : $I_3 \dot{\omega}_3 = 0 \Rightarrow \dot{\omega}_3 = 0$ or $\omega_3(t) = \text{const}$

$\rightarrow$ From first two equations

$$\left. \begin{aligned} \dot{\omega}_1 &= - \left(\frac{I_3 - I_1}{I_1} \omega_3 \right) \omega_2 \\ \dot{\omega}_2 &= \left(\frac{I_3 - I_1}{I_1} \omega_3 \right) \omega_1 \end{aligned} \right\} \quad \begin{aligned} &\text{Let us define} \\ &\Omega = \frac{I_3 - I_1}{I_1} \omega_3 \end{aligned}$$

so that

$$\left. \begin{aligned} \dot{\omega}_1 + \Omega \omega_2 &= 0 \\ \dot{\omega}_2 - \Omega \omega_1 &= 0 \end{aligned} \right\} \quad \text{coupled equations}$$

Derivate the first one: $\ddot{\omega}_1 + \Omega \dot{\omega}_2 = 0 \rightarrow \dot{\omega}_2 = -\ddot{\omega}_1 / \Omega$

Plug the result into the second: $-\frac{\ddot{\omega}_1}{\Omega} - \Omega \omega_1 = 0 \rightarrow \boxed{\ddot{\omega}_1 + \Omega^2 \omega_1 = 0}$
Harmonic osc. eq. of ω_1

$$\Rightarrow \boxed{\omega_1(t) = \omega_0 \cos(\Omega t)} \rightarrow \dot{\omega}_1(t) = -\omega_0 \Omega \sin(\Omega t)$$

$$\text{By using the first eq: } -\omega_0 \Omega \sin(\Omega t) + \Omega \omega_2 = 0 \Rightarrow \boxed{\omega_2(t) = \omega_0 \sin(\Omega t)}$$

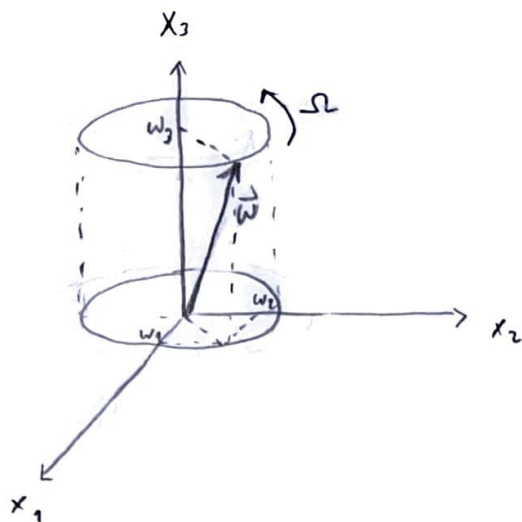
Therefore:

$$\boxed{\begin{aligned} \omega_1(t) &= \omega_0 \cos(\Omega t) \\ \omega_2(t) &= \omega_0 \sin(\Omega t) \end{aligned}}$$

we know that ω_3 : constant

$$|\vec{\omega}| = \omega = \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2} = \sqrt{\omega_0^2 + \omega_3^2} : \text{constant}$$

$\rightarrow$ parametric equations of a circle.



★ x_3 is the symmetry axis of the body.
So, $\vec{w}$ revolves and precesses about the body x_3 axis with a constant angular frequency Ω .

★ Thus, to an observer in the body coordinate system, $\vec{w}$ traces out a cone around the body symmetry axis, called the "body cone".

CONSTANT OF MOTION #1

★ We are considering a force-free motion.
So $\vec{L}$ is constant in the fixed coordinate system.

CONSTANT OF MOTION #2

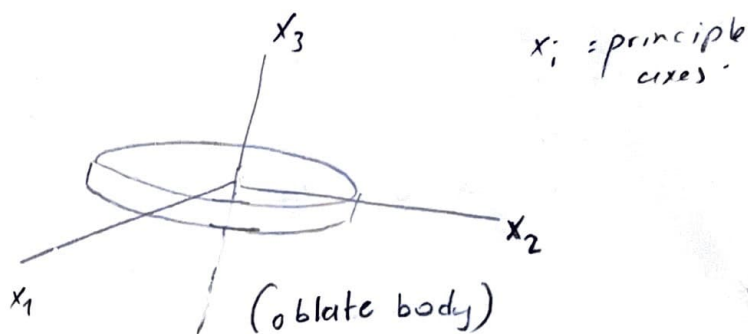
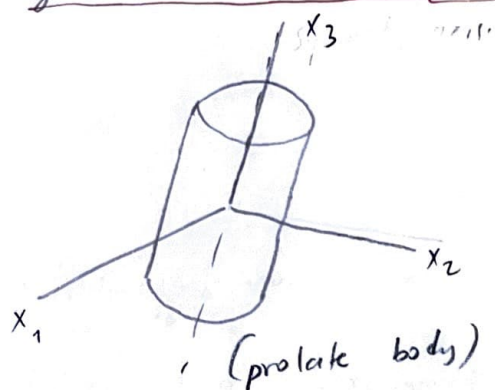
★ Body's COM is fixed and no torque applied on the body.

$$T_{\text{rot}} = \frac{1}{2} \vec{w} \cdot \vec{L} = \text{constant.}$$

★ We also know that $\vec{L}$ is constant. So $\vec{w}$ must move such that its projection on the stationary angular-momentum vector is constant.
Thus, $\vec{w}$ precesses around $\vec{L}$ making a constant angle with the vector $\vec{L}$. In such a case $\vec{L}$, $\vec{w}$ and x_3 (body) axis all lie in a plane.

Symmetric top (No torque)

- We would like to know how a symmetric top, i.e., an object like a cylinder or a disc can rotate in the space in the most general case if there is no force or torque acting on it.



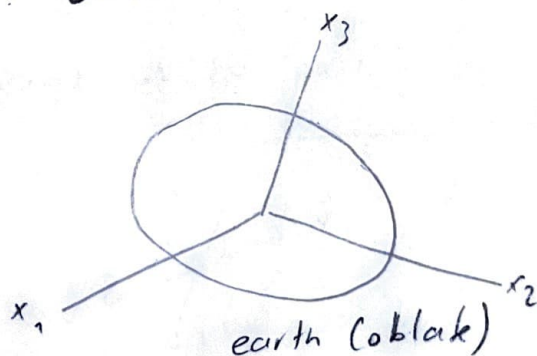
x_i = principle axes.

$$I_1 = I_2 \neq I_3 : \text{symmetric top.}$$

I_3 : Unique moment, i.e., it is different than the others.

- If I_3 is the largest of three, then the body is an "oblate", if it is the smallest the body is "prolate".

For example: earth is an oblate body:



- Euler's equations give the relationship between the torque and angular momentum

$$I_1 \dot{\omega}_1 - (I_1 - I_3) \omega_2 \omega_3 = N_1$$

$$I_2 \dot{\omega}_2 - (I_3 - I_1) \omega_3 \omega_1 = N_2$$

$$I_3 \dot{\omega}_3 - (I_1 - I_2) \omega_1 \omega_2 = N_3$$

equations are

→ Please keep in mind that all quantities in these [^]defined for x_i (body coordinates); Euler's equations are written in body coordinates.

★ Once again, note that we are interested in the most general rotation of a symmetric top could possibly do. otherwise, think that we can simply, for example, rotate a prolate body about one of its symmetry axes:



→ and it will be rotating about this axis forever with an angular momentum $\vec{L}$ in the same direction, with x_3 and $\vec{w}$:

$$\vec{w} = w_3 \hat{e}_3$$

$$\vec{L} = L_3 \hat{e}_3$$

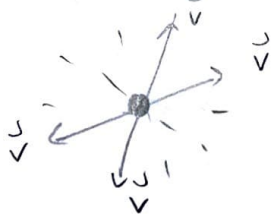
? Is this the most general rotation that a prolate body can do?

★ Think of a point particle:

$$\vec{F} = \frac{d\vec{p}}{dt} \quad \text{if there is no force acting on it} \rightarrow 0 = \frac{d\vec{p}}{dt} \Rightarrow \vec{p} = \text{constant.}$$

If we solve $\vec{F} = \frac{d\vec{p}}{dt}$ for $\vec{F} = 0$ we obtain the general solution $\vec{p} = \text{constant} \Rightarrow m\vec{v} = \text{constant}$

$\Rightarrow \vec{v} = \text{constant} \Rightarrow$ the particle can move in any direction in the space with a constant velocity.



So, solving $\vec{F} = d\vec{p}/dt$ for $\vec{F} = 0$ provided us the most general solution for the velocity of the point particle.

★ Similar to above example, if we set $\vec{N} = 0$ for Euler's equations, we will obtain the most general solution for $\vec{w}$, and accordingly, $\vec{L}$; that is to say we will know about how a rigid body can rotate in the space in the most general case.

For example imagine spinning a dinner plate in the air. Why does it rotate in the same wobbly way regardless how we throw it?

(Newton's 2nd law for rotations.)

When we set $N_1 = N_2 = N_3 = 0$, Euler's equations become,

$$\dot{\omega}_1 = \frac{I_1 - I_3}{I_1} \omega_3 \omega_2$$

$$\dot{\omega}_2 = \frac{I_3 - I_1}{I_2} \omega_3 \omega_1$$

$$\dot{\omega}_3 = \frac{I_1 - I_2}{I_3} \omega_1 \omega_2$$

Since we are considering a symmetric top $I_1 = I_2 \neq I_3$ we can simply replace I_2 's with I_1 (since they are the same) - Then, the Euler's equations become (with some other small arrangements)

$$\dot{\omega}_1 = - \left[\frac{I_3 - I_1}{I_1} \omega_3 \right] \omega_2$$

$$\dot{\omega}_2 = \left[\frac{I_3 - I_1}{I_1} \omega_3 \right] \omega_1$$

$$\dot{\omega}_3 = 0 \rightarrow \text{it is straightforward to solve this equation} \rightarrow \omega_3 = \text{constant}$$

Since I_1, I_3 and ω_3 are constant, let's denote $\frac{I_3 - I_1}{I_1} \omega_3$ by

$$\Omega = \frac{I_3 - I_1}{I_1} \omega_3$$

Then Euler's equations become

$$\begin{cases} \dot{\omega}_1 + \Omega \omega_2 = 0 \\ \dot{\omega}_2 - \Omega \omega_1 = 0 \end{cases}$$

These are called coupled differential equations, i.e., ω_1 appears with a derivative in one of the equations and without a derivative in the other one. (This is true for ω_2)

To "decouple" these equations, we can take the time derivative of the first equation,

$$\ddot{\omega}_1 + \Omega \dot{\omega}_2 = 0$$

and plug $\dot{\omega}_2$ from the second eq. to the first one:

$$\ddot{\omega}_1 + \Omega^2 \omega_1 = 0$$

$\ddot{w}_1 + \Omega^2 w_1 = 0$ is the equation of motion for the simple harmonic oscillator and we know its solution:

$$w_1(t) = w_0 \cos(\Omega t + \delta)$$

We can find $w_2(t)$ by using this solution;

$$\dot{w}_1(t) = -\Omega w_0 \sin(\Omega t + \delta)$$

Plug this in $\dot{w}_1 + \Omega w_2 = 0$:

$$-\cancel{\Omega} w_0 \sin(\Omega t + \delta) + \cancel{\Omega} w_2 = 0$$

$$w_2(t) = w_0 \sin(\Omega t + \delta)$$

Therefore by solving Euler equations we have found the following results for the angular momentum:

$$\begin{aligned} w_1(t) &= w_0 \cos(\Omega t + \delta) \\ w_2(t) &= w_0 \sin(\Omega t + \delta) \\ w_3(t) &= \text{constant} \end{aligned}$$

→ The components of the angular velocity in body coordinates.

What does these equations tell us?

- (i) $\hat{e}_3$ component of the angular velocity vector must be constant in time according to $w_3(t) = \text{constant}$
- (ii) $w_1(t)$ and $w_2(t)$ give us information about how the projection of $\vec{w}$ vector in $\hat{e}_1 - \hat{e}_2$ plane behave:

$$w_1^2 = w_0^2 \cos^2(\Omega t + \delta)$$

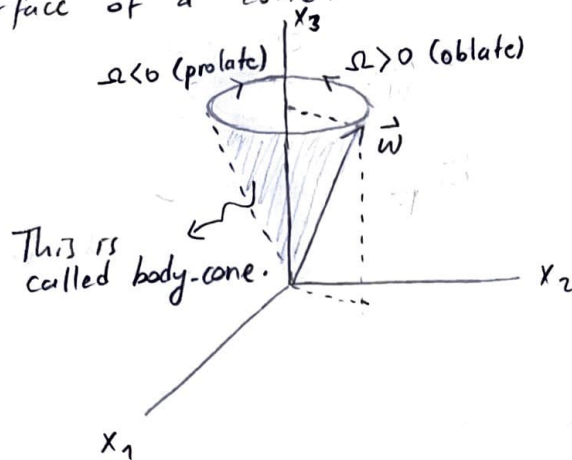
$$w_2^2 = w_0^2 \sin^2(\Omega t + \delta)$$

$$w_1^2 + w_2^2 = w_0^2 \left[\underbrace{\cos^2(\Omega t + \delta) + \sin^2(\Omega t + \delta)}_1 \right]$$

$$w_1^2 + w_2^2 = w_0^2 : \text{equation of circle.}$$

⇒ So, the projection of $\vec{w}$ in $\hat{e}_1 - \hat{e}_2$ follows a circular path with radius w_0 .

- Therefore, if we combine results (i) and (ii) we conclude that the angular velocity vector rotates around $\hat{e}_3$ with an angular speed Ω sweeping the surface of a cone:



Ω : angular speed of $\vec{w}$ around x_3 (or $\hat{e}_3$)

$$\Omega = \frac{I_3 - I_1}{I_1} \omega_3$$

- Until now we have just dealt with how instantaneous axis of rotation $\vec{w}$ precesses about the body-fixed symmetry axis x_3 describing the body cone. This is not the only motion that symmetric-top does.

- $I_3 > I_1$: the top is oblate

$$\Rightarrow \Omega > 0$$

- $I_1 > I_3$: the top is prolate

$$\Rightarrow \Omega < 0$$

- Think of the angular momentum vector $\vec{L}$ from the point of view of the fixed-coordinate axes x_i' . Since there is no net torque acting on the body $\vec{L}$ must be constant in x_i' .
- On the other hand, ^{since} the rotating body is not under the influence of any force or torque, the total kinetic energy of the body must be constant. Also, ^{we assume that} the C.O.M. of the body does not move with respect to the fixed coordinate axes. The total kinetic energy of the body is only the rotational K.E. of the body which is constant.

$$T_{\text{rot}} = \frac{1}{2} \vec{w} \cdot \vec{L} = \text{constant} (*)$$

$\Rightarrow$ In x_i' $\vec{L}$ is fixed. $\Rightarrow$ According to (*) $\vec{w}$ must move with respect to $\vec{L}$ such that its projection on $\vec{L}$ is constant. $\Rightarrow$ $\vec{w}$ must precess around $\vec{L}$ and the angle between $\vec{w}$ and $\vec{L}$ must be constant.

- In summary, $\vec{w}$ rotates about x_3 , and it also rotates about $\vec{L}$ in x_i' . So, how does the body move in x_i' frame all in all? To understand this, we need to consider one more thing:

- Let us show that $\vec{L}$, $\vec{\omega}$ and $\hat{e}_3$ (x_3 axis) all lie in a plane.

To show this it is enough to see that the following identity holds:

$$\vec{L} \cdot (\vec{\omega} \times \hat{e}_3) = 0$$

Why? The result of the cross product $\vec{\omega} \times \hat{e}_3$ is perpendicular to the plane where $\vec{\omega}$ and $\hat{e}_3$ lie, i.e., the vector $\vec{\omega} \times \hat{e}_3$ is perpendicular to both $\vec{\omega}$ and $\hat{e}_3$. If $\vec{L}$ has no projection on $\vec{\omega} \times \hat{e}_3$ (which means $\vec{L} \cdot (\vec{\omega} \times \hat{e}_3) = 0$) then $\vec{L}$ have to be perpendicular to $\vec{\omega} \times \hat{e}_3$. This necessarily requires $\vec{L}$ to lie in the same plane with $\vec{\omega}$ and $\hat{e}_3$ lie.

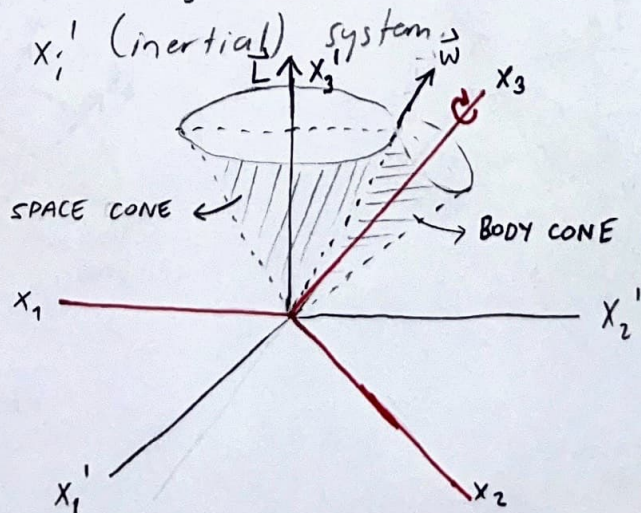
$\Rightarrow \vec{L}$, $\vec{\omega}$ and $\hat{e}_3$ all lie in a plane!

- To prove $\vec{L} \cdot (\vec{\omega} \times \hat{e}_3) = 0$:

$$\vec{\omega} \times \hat{e}_3 = (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) \times \hat{e}_3 = -\omega_1 \hat{e}_2 + \omega_2 \hat{e}_1$$

$$\begin{aligned} \vec{L} \cdot (\omega_2 \hat{e}_1 - \omega_1 \hat{e}_2) &= L_1 \omega_2 - L_2 \omega_1 = I_1 \omega_1 \omega_2 - I_2 \omega_1 \omega_2 \\ &= (I_1 - I_2) \omega_1 \omega_2 = 0 \quad \text{since } I_1 = I_2 \end{aligned}$$

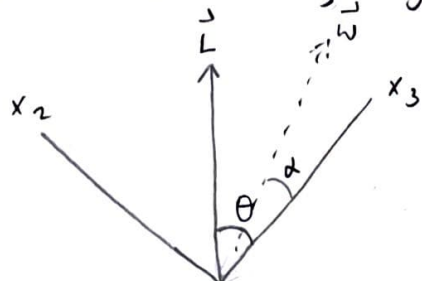
- How can we describe the motion pictorially? Let us choose x_3' to coincide with $\vec{L}$. Then, an observer in x_i' system will see $\vec{\omega}$ traces out a cone around x_3' axis called space cone. Since $\vec{\omega}$ also traces out the body cone at the same time and we have already shown that $\vec{L}$, $\vec{\omega}$ and x_3' lie in the same plane, the body cone have to roll on the space cone such that $\vec{\omega}$ precesses around the x_3 -axis in the body system and around x_3' -axis (or $\vec{L}$) in the x_i' (inertial) system.



- Body rotates about its symmetry axis x_3 and x_3 rotates about x_3' ($\vec{L}$). This is the most general rotation that a body can do.

- Actually, the motion shown in the previous figure actually refers to the motion of a prolate object ($I_1 > I_3$). For an oblate object the space cone would be inside the body cone rather than outside. Let us prove these facts.

→ The key idea is that since x_1 - x_2 plane rotates about x_3 axis, at some instant x_2 axis will be in the plane defined by $\vec{L}$, $\vec{\omega}$ and $\hat{e}_3$. If we denote the angle between $\vec{L}$ and $\hat{e}_3$ by θ , then



$$L_1 = 0 \quad (x_1 \text{ is perpendicular to the plane where } \vec{L}, \vec{\omega}, x_3 \text{ and } x_2 \text{ lie})$$

$$L_2 = L \sin \theta$$

$$L_3 = L \cos \theta$$

→ Let the angle between $\vec{\omega}$ and x_3 be α . Then, at the instant indicated above

$$\omega_1 = 0$$

$$\omega_2 = \omega \sin \alpha$$

$$\omega_3 = \omega \cos \alpha$$

→ we can determine the components of $\vec{L}$ from $L_i = I_i \omega_i$:

$$L_1 = I_1 \omega_1 = 0$$

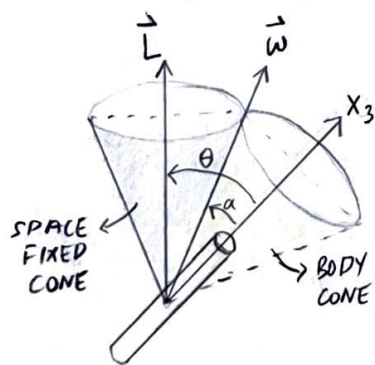
$$L_2 = I_2 \omega_2 = I_2 \omega \sin \alpha$$

$$L_3 = I_3 \omega_3 = I_3 \omega \cos \alpha$$

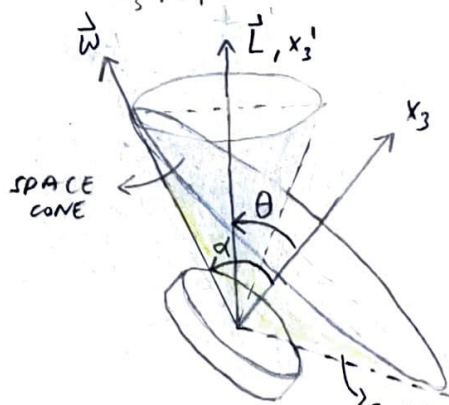
→ So, we can calculate the ratio L_2/L_3 :

$$\frac{L_2}{L_3} = \frac{I_2}{I_3} \tan \alpha = \tan \theta$$

$$\Rightarrow \begin{aligned} I_1 > I_3 &\rightarrow \theta > \alpha && \text{prolate} \\ I_3 > I_1 &\rightarrow \alpha > \theta && \text{oblate} \end{aligned}$$



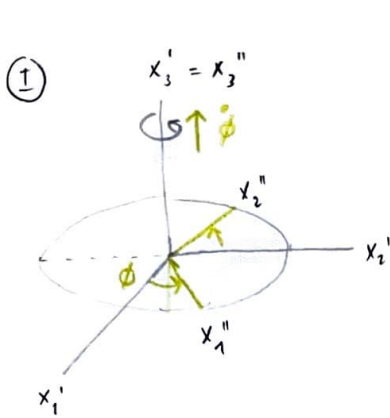
OBLATE BODY



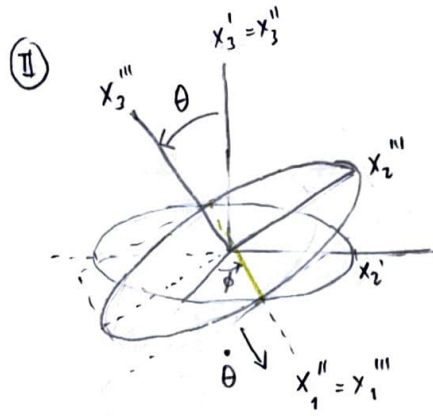
PROLATE BODY

Eulerian Angles:

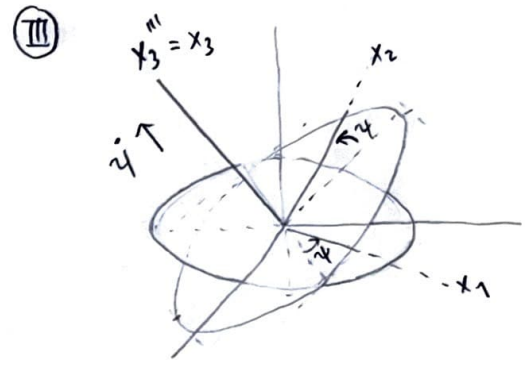
We want to transform from the fixed coordinates x_i' to body coordinates x_i .



ϕ about x_3'
counterclockwise



θ about x_1''
counterclockwise



ψ about x_3'''
counterclockwise

- Any vector described in x_i' basis: $\vec{x}'$
- Any vector described in x_i basis: $\vec{x}$

Transformation from $x_i' \rightarrow x_i$

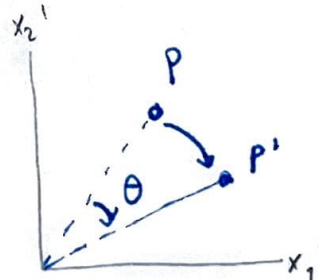
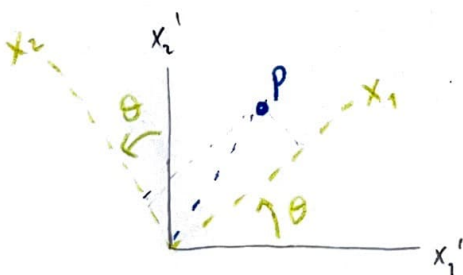
$$\vec{x} = [\vec{\lambda}] \vec{x}'$$

Transformation matrix

$[\vec{\lambda}]$ consists 3 separate rotation stated as above

$$[\vec{\lambda}] = [\vec{\lambda}_\psi][\vec{\lambda}_\theta][\vec{\lambda}_\phi] \quad \phi, \theta, \psi \text{ are called EULERIAN ANGLES.}$$

How do we describe a rotation on a plane?

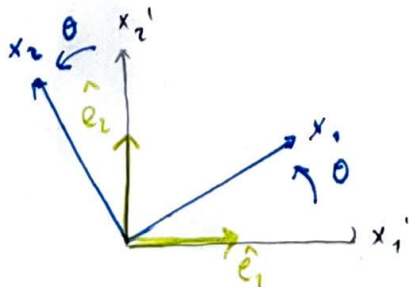


Two equivalent approach.

Assume that we have the unit vectors in x_i' system $\hat{e}_i'$. So we can describe any vector as

$$\vec{v} = v_1 \hat{e}_1' + v_2 \hat{e}_2'$$

How can we describe $\hat{e}_1$ and $\hat{e}_2$ in a rotated coordinate system?



$$\hat{e}_1' = \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix} \rightarrow \begin{matrix} x_1 \text{ component} \\ x_2 \text{ component} \end{matrix}$$

$$\hat{e}_2' = \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$$

$$\hat{e}_1' = \cos \theta \hat{e}_1 - \sin \theta \hat{e}_2$$

$$\hat{e}_2' = \sin \theta \hat{e}_1 + \cos \theta \hat{e}_2$$

$$\vec{v} = v_1 (\cos \theta \hat{e}_1 - \sin \theta \hat{e}_2) + v_2 (\sin \theta \hat{e}_1 + \cos \theta \hat{e}_2)$$

$$= (v_1 \cos \theta + v_2 \sin \theta) \hat{e}_1 + (-v_1 \sin \theta + v_2 \cos \theta) \hat{e}_2$$

To transform the components $\rightarrow \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

rotation matrix

- ϕ rotation takes place in the $x_1' - x_2'$ plane:

$$[\vec{\lambda}_\phi] = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- θ rotation takes place in $x_2'' - x_3''$ plane

$$[\vec{\lambda}_\theta] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

- γ rotation takes place in $x_1''' - x_2'''$ plane:

$$[\vec{\lambda}_\gamma] = \begin{bmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

